Credit Models for Quants

Pricing, Default Risk, Correlation & PnL Across Credit Products
Why Desks Choose Intensity, Copula, LSS — and When They Break

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Table 4: Credit Regime Recognition & Model Selection Guide

Credit Type	Examples	Typical Spread Skew	Default Clustering	Primary Model	Red Flag
IG Financials	JPM, GS CDS	Flat (50-150bp)	Low	Intensity + low	Ignoring wrong-way risk
HY Corporates	TSLA, F (post-2012)	Steep (500-2000bp)	Medium	Jump-to-Default + (t)	Using constant
Sovereign EM	Turkey, Argentina	Extreme (>2000bp)	High	Intensity + jump process	Assuming LGD = 40%
Distressed	Lehman 2008, Evergrande	Inverted (spread < 0)	Certain	Recovery + JTD	Using diffusion models
CDO Tranches	CDX.IG 0-3%	Correlation smile	Very High	Local Correlation	Gaussian copula + static
Loan CDS	LCDX	Higher spreads vs bonds	Medium	Intensity + prepayment	Assuming R = bond R

Quick Recognition Guide

Interview Trick - 3 Questions to Classify Any Credit:

1. **Is it a financial?** (JPM, GS) → Expect wrong-way risk with rates, use hybrid
2. **Is it distressed?** (spread > 1000bp) → Default is certain, use JTD + recovery swap
3. **Is it tranched?** (0-3% equity) → Correlation is all that matters, use local

Table 5: Crisis Mode Adjustments - When Normal Models Break

Crisis Type	What Breaks	Model Adjustment	Reserve Multiplier	Example
Default Clustering	Independence of across names	Multiply by sector factor (2-5×)	3-5× normal	2008 Financials collapse
Correlation Explosion	Gaussian copula static	Local correlation: (L) → 1 at attachment	5-10× normal	2005 Correlation crisis
Recovery Lock	LGD assumed constant (40%)	Price recovery as stochastic, hold recovery swaps	10× normal	Lehman (R 10% vs 60% assumed)
Liquidity Freeze	CDS bid-ask 10× normal	Widen CS01 by 3×, hold extra JTD reserves	3× normal	COVID March 2020
Wrong-Way Risk	Credit × Rates/Fx correlation ignored	Add correlation term to drift, use hybrid model	5× normal	DB 2016 (rates up = credit down)
Peg Break (Sovereign)	Diffusion model for pegged currency	Binary jump model, hold 30% notional as reserve	10× normal	CHF 2015 (sovereign)
Jump-to-Default Event	model assumes smooth default	JTD = 100% notional loss, no hedge	Cannot hedge	Evergrande 2021

The 3-Way Credit Distinction

Asset	Drift	Loss	Hedge
Equity	$\mu > 0$	Continuous	Delta
FX	$r_d - r_f$	Two-sided	Forward
Credit	None	Jump (binary)	Reserve

$$\boxed{\text{CDS} = \frac{\text{Loss}}{\text{Time}} = \lambda \times \text{LGD}}$$

Module 2: Reduced-Form (Intensity) Models—Default as a Clock

Module Overview

What this module is about: This module introduces reduced-form models where default arrives via a hazard rate $\lambda(t)$, not from an asset crossing a barrier.
Why this matters: Intensity models are the workhorse of credit trading floors. They calibrate directly to CDS and allow fast pricing.
What confusion this clears:

- Where comes from (market-implied, not historical)
- Why intensity models dominate flow books
- Why is piecewise-constant in practice

After this module, you should be able to answer: "Why do desks use intensity models even though they ignore the economic story of default?"

2.1 Intensity SDE Survival Probability

SDE for default indicator:

Core Formula

$$dN_t = \begin{cases} 1 & \text{with probability } \lambda_t dt \\ 0 & \text{with probability } 1 - \lambda_t dt \end{cases}$$

Survival probability:

Core Formula

$$\mathbb{P}(\tau > T) = \mathbb{E}\left[e^{-\int_0^T \lambda(s) ds}\right]$$

In practice: (t) is assumed deterministic or piecewise constant:

Core Formula

$$\lambda(t) = \sum_{i=1}^{n} \lambda_i \mathbb{1}_{[T_{i-1}, T_i)}(t)$$

Practitioner Insight

The Correlation Smile:
Just like FX vol smile, credit has correlation smile.
Market quotes (2007): - 0-3% tranche: $\rho = 0.15$ - 3-7% tranche: $\rho = 0.25$ - 7-10% tranche: $\rho = 0.35$ - 10-15% tranche: $\rho = 0.45$
What this means: - Equity tranche (0-3%) is **long correlation** - Senior tranche (10-15%) is **short correlation** - Mezz tranche (3-7%) is **neutral-ish**
The Trade: - If you think correlation will increase → buy equity tranche, sell senior - If you think correlation will decrease → sell equity, buy senior
Why It Failed in 2008: - Correlation went to 1.0 across all tranches - Equity tranche lost (too much correlation) - Senior tranche lost (defaults exceeded attachment) - All tranches lost simultaneously → model was wrong
Interview Gold: "Base correlation is not a model—it's a quoting convention. It's like GK vol in FX. It tells you where the market is, but not where it's going. And like GK, it breaks when you need it most."

$\boxed{\rho_{\text{crisis}} = \rho_{\text{normal}} + \Delta\rho \cdot \mathbb{1}_{\{\text{VIX}>30\}}}$ Where: $\rho_{\text{normal}} \approx 0.3, \quad \Delta\rho \approx 0.6$

Tranche	$\partial P / \partial \rho$	Sign
0-3% Equity	Large	Positive (long ρ)
3-7% Mezz	Medium	Near zero
7-10% Senior	Large	Negative (short ρ)

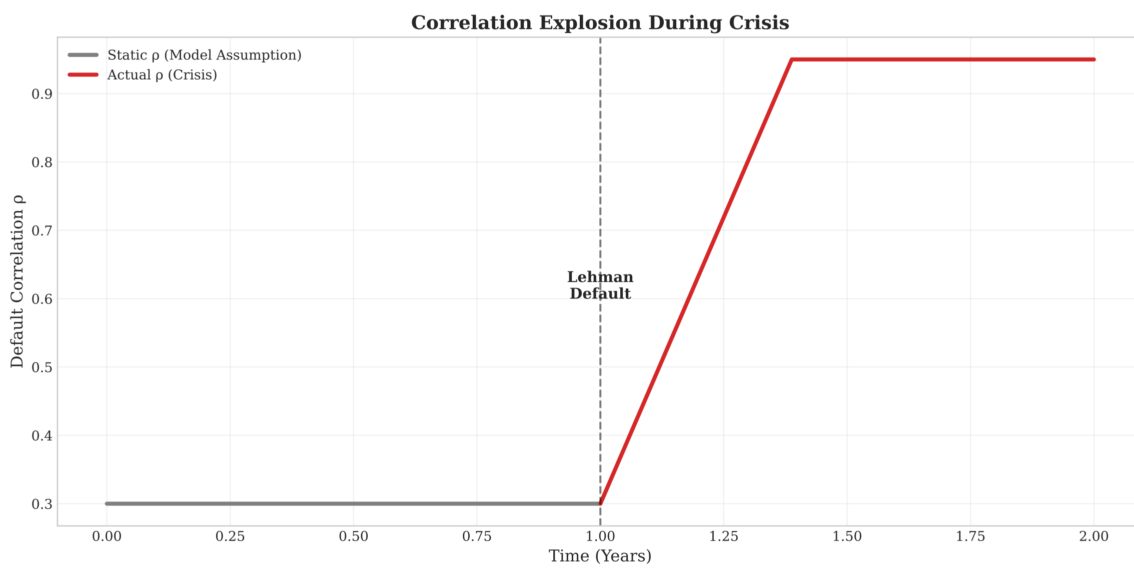

Figure 7: Correlation evolution during crisis: static fails to capture regime shift from 0.3 to 0.95, causing simultaneous losses across all tranches

Correlation is a Smoke Alarm, Not a Thermostat

Interview anchor: "Correlation doesn't gradually increase—it jumps to 1 when systemic risk hits. Gaussian copula models it as a thermostat (gradual adjustment). Reality is a smoke alarm (binary trigger)."
Desk Wisdom: - Monitor correlation skew daily: if 0-3% tranche jumps >0.05 in a day, systemic fear is building - In crisis, correlation vega becomes infinite—model breaks down - The only hedge: cut position size by 50% when (0-3%) > 0.20
Memory Trick: " = 0.3 = independent defaults = normal times" " = 0.95 = all defaults together = crisis times" "Model only knows 0.3 → you're dead when it jumps"